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How does land consolidation drive rural industrial development? Qualitative and quantitative analysis of 32 land consolidation cases in China

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ABSTRACT

As rural revitalization becomes an increasingly central goal in countries worldwide, land consolidation has emerged as an important tool for realizing rural prosperity through facilitating industrial development. However, it is still not clear how and to what extent land consolidation contributes to rural industrial development. This study, therefore, investigates the influencing mechanism of factors related to land consolidation on rural industrial development. Following the element-structure-function framework, factors related to land consolidation are classified as follows: technical and public elements (e.g., public involvement), financing and organizational structures, production, living, and ecological functions. Cross-case studies are conducted using two emerging methods of causal complexity analysis, i.e., qualitative comparative analysis (QCA) and necessary conditions analysis (NCA). First, the results of QCA reveal three configurations of factors explaining rural industrial development, including function-driven configuration, function-element-driven configuration, and function-structure-driven configuration. Second, the results of NCA indicate the nuanced roles played by different factors at various levels of industrial development. Specifically, at the middle level of rural industrial development (50% quantile), production function becomes the necessary condition; at the upper-middle level of rural industrial development (between 50% and 80% quantile), technical element, financing structure, organizational structure, and living function become the necessary conditions for rural industrial development; public element becomes a necessary condition to achieve a high level of industrial development (80% quantile). This study shows how different configurations of factors related to land consolidation influence rural industrial development and the diversified roles that these factors play. The findings enable an extension to the influencing mechanism of land consolidation on rural revitalization and shed new light on formulating and aligning targeted land consolidation policies (e.g., endogenous incentives, crowdsourcing platform, and ecological development) to achieve superior industrial development in different rural areas.

1. Introduction

The Sustainable Development Goals of United Nations call for the eradication of poverty (Johnston, 2016). The global poverty rate in 2019 is 8.2%, and the projected poverty rate for 2020 is 8.6% (Buheji et al., 2020). Currently, the poverty challenge is largely rooted in rural areas (Zhou et al., 2019), especially in developing countries, such as China (W. Wang et al., 2021). To pursue the goal of ending poverty, governments formulate robust and flexible measures for sustainable development in rural areas, among which land consolidation is a pragmatic and powerful approach. Land consolidation refers to the redistribution of land parcels to enable a series of measures to increase economic activity and efficiency (e.g., agricultural land adjustment, rural infrastructure development, abandoned industrial and mining land reclamation, and rural landscape improvement) (Demetriou et al., 2012; Jiang et al., 2022; Munnangi et al., 2020). In a rural recession, land consolidation can play a crucial role in facilitating regional sustainable development by replenishing cultivated land, revitalizing the stock of land,

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strengthening intensive land use, and improving land production capacity (Li et al., 2018; Long, 2014; Long and Qu, 2018; Tu and Long, 2017). In particular, land consolidation has achieved great success in supporting rural industrial development (e.g., the added value of agriculture, rural tourism, and land productivity) in recent years (Li et al., 2018; Rao, 2022). As industrial development changes, the aim of land consolidation is also evolving (Zhou et al., 2020): improving the land system (Burton and King, 1982; Demetriou, 2014; Demetriou et al., 2012; Pašakarnis and Maliene, 2010; Uyan et al., 2015) as well as, more importantly, acting as a platform to promote rural revitalization (Demetriou et al., 2012; Du et al., 2018; Shi et al., 2018).

Land consolidation exhibits three types of factors, i.e., element, structure, and function, that shape or constrain its effects on rural industrial development. The diversified interaction of material and immaterial elements in China's rural areas, has shaped the production, living, and ecological functions, resulting in great changes in the structures of industry, employment, consumption, land use, and social organization in rural areas (Li et al., 2015; Long et al., 2016, 2012, 2011). First, the element relates to resource factors, such as public element and technological element. The public element refers to the willingness to participate and the degree of participation among farmers in the process of land consolidation. Louwsma et al. (2014) identified the importance of public involvement for maximizing the benefits of land consolidation. The technological element refers the application of feasible, suitable, and advanced farming technologies for land consolidation. The level of available technology has a significant impact on production efficiency (Yao et al., 2016). Second, the structure involves institutional factors, such as financing structure and organizational structure. The financing structure refers to the financing arrangements of land consolidation. Public-private partnership (PPP) is an increasingly prevalent financing arrangement in land consolidation, and its effect on poverty reduction is significant (W. Wang et al., 2021). The organizational structure refers to the organization of professional cooperatives or enterprises in implementing land consolidation. Different organizational models produce various effects on agricultural ecological production (Xie et al., 2021). Third, the function of land manifests the direct effects of land consolidation, which alters the spatial patterns of production, life, and ecology (Long, 2014). According to Munnangi et al. (2020), the function of land consolidation extends from cultivated land production promotion to comprehensive improvement of agricultural production conditions, rural settlement, and ecosystems. More specifically, land consolidation contributes gradually to agricultural scale production, concentrated rural residence, and ecological transformation (Shan et al., 2019; Zhou et al., 2020).

In sum, land consolidation is a systematic project involving element-related, structure-related, and function-related factors. These factors are interdependent and work together to shape outcomes in land consolidation. However, there is a research gap between land consolidation and industrial development in rural areas. First, earlier studies present multiple qualitative analyses of the relationship between land consolidation and rural industrial development (Jiang et al., 2021; Pašakarnis and Maliene, 2010; Rao, 2022; Zhou et al., 2019, 2020). However, quantitative evidence regarding this relationship is still lacking. It remains unclear to what extent land consolidation contributes to rural industrial development. Second, although several recent studies have conducted quantitative analyses of the effect that land consolidation yields on agricultural productivity and income growth (Du et al., 2018; W. Wang et al., 2021; Wojewodzic et al., 2021), limited quantitative research has been conducted on the comprehensive effects of land consolidation on rural industrial development. Third, prior studies mainly focused on certain factors of land consolidation, such as public involvement (Krupowicz et al., 2020), technology application (Yao et al., 2016), organization model (Haldrup, 2015), and functional layout (Long, 2014). A systematic approach that incorporates element-related, structure-related, and function-related factors seems to be missing. In sum, the effects of land consolidation on rural industrial development are complex and hinge on interdependent factors, requiring an in-depth exploration. Following the route laid out by Fainshmidt et al. (2020), this study employs and combines two emerging methods, i.e., qualitative comparative analysis (QCA) and necessary conditions analysis (NCA), to systematically address the following two research questions (RQ):

RQ1:. *How do different configurations of land consolidation related factors (i.e., element-structure-function factors) influence rural industrial development?*

RQ2:. *To what extent do land consolidation related factors (i.e., element-structure-function factors) constrain rural industrial development?*

The remainder of this article proceeds as follows. Section 2 reviews the connotations of land consolidation and its driving mechanisms for industrial development. Section 3 presents the research methods, the collection of cases, and the measurement of land consolidation (i.e., element-structure-function factors) and industrial development (i.e., outcome). The results are presented in Section 4, including the configurations of factors related to land consolidation and the bottleneck of these factors. Section 5 discusses the implications of QCA and NCA analyses, respectively. Finally, Section 6 provides concluding remarks.

2. Literature review and theoretical framework

Over the past decade, a growing number of studies have focused on the connotation of land consolidation and its influencing mechanism on rural industrial development. The relevant literature is, therefore, analyzed in relation to the following two aspects.

2.1. Land consolidation perspectives

Land consolidation is a worldwide phenomenon. It is useful for improving land productivity and land use effectiveness (Jiang et al., 2022), rationalizing social and economic issues in managing the development of rural areas (Bonadonna et al., 2020; Wójcik-Leń et al., 2018), as well as dealing with nature conservation and environmental issues (Moravcová et al., 2017). As an integrated concept (Jiang et al., 2022), land consolidation incorporates land reallocation (Cay and Iscan, 2011), land readjustment (Muñoz Gielen and Mualam, 2019), land rearrangement (Xu et al., 2017), and land reclamation (Cao, 2007). The concept of consolidation originated in Europe, and its development has passed through several stages. In the 1960 s and 1970 s, most European countries established land consolidation projects to stimulate the economic and social effects, bring about a better balance between rural and urban areas, and resolve land-use conflicts (Jiang et al., 2022). A quest for environmental integration and public participation is proposed in many countries (Jiang et al., 2022; Louwsma et al., 2014). Increasing research has been conducted to ensure the overall goal of economic, social, and ecological benefits (Bonadonna et al., 2020; Jiang et al., 2022; Wójcik-Leń et al., 2018). In China, land consolidation involves the rational organization of the adjustment of land utilization, contributing to the improvement of production efficiency and comprehensive rural development. According to Long et al. (2019), the conception of land consolidation in China has undergone three stages of evolution, switching focus from compensation for the loss of cultivated land to integrative consolidation of cultivated land and rural settlements and to the restoration of land ecological functions. Since the reform and opening up period began in 1978, the rapid development of urbanization has redued the amount of cultivated land in China. This has exacerbated tensions regarding insufficiencient cultivated land resources (Liu et al., 2014). Since the mid-1990 s, China has increased its amount of cultivated land, in particular utilizing land consolidation (Guanghui et al., 2015).

Theoretically, land consolidation is a complex system that integrates elements, structures, and functions (Li et al., 2018). Public and technical elements, optimized financing and organizational structures, and the spatial restructuring of the land for production, living, and ecological functions are involved in the implementation of land consolidation (Zhou et al., 2020). Thus, systematic thinking, which is the theoretical lens of studying the structure and function of a system, is indispensable for comprehending the land consolidation system. A system is an organic whole with certain functions, consisting of several elements embedded into a certain structural form (Myers, 2003). More specifically, elements reflect the resources of the system; structure is the institutional settings of the system; and function captures the effects of the system (Arnold and Wade, 2015). System thinking contributes to the development of a holistic understanding of complex systems in the realm of land management (Church et al., 2020; Mai et al., 2019; Nguyen et al., 2017). The goal of system thinking is not only to develop a detailed understanding of each factor related to the system but, more importantly, to characterize the system as a whole. The success of a land consolidation project hinges on the configuration of several types of factors (Zhou et al., 2020). It is, therefore, important to conduct a systematic analysis of the factors related to land consolidation. Drawing upon the system thinking, element-structure-function provides a holistic framework for unraveling the factors related to land consolidation (Demetriou et al., 2012; Liu et al., 2012; Long et al., 2019) and can be used to examine the complex role that land consolidation plays in facilitating sustainable rural development (Long et al., 2019).

First, element involves resource factors, such as public involvement and technical introduction. Land consolidation can facilitate positive interactions between people and land across several stages, as well as introducing advanced engineering and ecological technologies (Long et al., 2019). The mobilization of public involvement in land consolidation is crucial (Krupowicz et al., 2020). Land consolidation, therefore, requires measures that facilitate the engagement of rural residents and stimulate their willingness and creativity. Public participation is needed at a series of stages (i.e., planning, implementation, completion and handover, and operation and maintenance) (Jiang et al., 2022), which improves the transparency of the consolidation process and enhances consolidation outcomes. In particular, public opinion should be collected during the planning stage to enable effective operation and maintenance (Xu et al., 2017). People are generally satisfied with the possibility of participation and this, in turn, enhances the efficiency of consolidation. On the other hand, as a comprehensive process, land consolidation requires the multidisciplinary integration of various resources. Technology is an important strategic resource in land consolidation. The introduction of advanced and applicable technologies is necessary for increase agricultural production. However, if the introduced technology does not fit the nature of the farming system, it may have a negative impact on agricultural production (Eastwood et al., 2010). A range of factors, such as the farming system, desired production function, and financing and profit, affect the adoption of technologies (Asiama et al., 2017).

Second, structure refers to institutional factors, such as organizational and financial structures. The financial channels of land consolidation are increasingly pluralistic, including government, enterprise, agricultural production cooperatives, and rural households (W. Wang et al., 2021). The diversified financial structures can boost the efficiency of land consolidation and the development of modern agriculture (Tang et al., 2017). Coupled with this trend, land consolidation projects are increasingly transitioning from entirely government investment to government-dominant investment and to market-dominant investment. In this process, organizational structures have evolved from top-down to bottom-up approaches. Although land consolidation projects may make use of investments outside of the government, such projects can continue to be led by the government. Thus, financial structures demonstrate the diversity of investments in general (i.e., governmental and market dependency), and organizational structures further reveal the power relationship between government and market (i.e., government and market power). Government-dominated land consolidation is a top-down approach focusing on land leveling and the construction of infrastructures (e.g., agricultural water conservancy facilities and field roads), while market-dominated land consolidation shows bottom-up features, focusing on facilitating modern agricultural development and exploiting the growth of profits.

Third, function refers to the direct effects of land consolidation through spatial reconfiguration of the land for production, living, and ecological functions (Long, 2014). Land-use transition, especially during urbanization, triggers huge changes in rural production and living spaces, as well as potentially creating severe ecological problems, which can be major impediments to sustainable rural development. Land consolidation provides an opportunity to realize the multifunctional cultivated land. Through comprehensive measures (relocation of farmers and the transformation of the rural landscape), land consolidation can effectively guide the concentration of fragmented cultivated land and enable large-scale production (Demetriou et al., 2012; Long, 2014). In addition to the production function, land consolidation can also contribute to preserving rural areas with poor living conditions through relocation and aggregation of their residences and improving mechanisms of ecological compensation (Zhou et al., 2019). Over the course of the spatial restructuring of production, living, and ecological functions, land consolidation is expected to drive rural industrial development, including organic, tourism, ecological, and cultural forms of agriculture (Long et al., 2019).

2.2. Influencing mechanism of land consolidation on industrial development

Rural revitalization refers to the improvement of the economy and social welfare, as well as the enhancement of ecological and cultural protection in rural areas. The core purpose of rural revitalization is establishing an interactive model that incorporates various factors, including population, technology, investment, and land, in a systematic manner (Zhou et al., 2020). Land consolidation involves the consolidation of cultivated land, construction land, and industrial and mining land, all of which require the restructuring of rural production, settlement, and ecological spaces through engineering and technical approaches (Jiang et al., 2022). Therefore, as a platform for integrating various factors, land consolidation plays a pivotal role in ensuring the high-quality development of rural industries (Kong et al., 2019). In particular, land consolidation, on one hand, improves agricultural productivity through reducing land fragmentation and expanding the development potential of the agricultural industry. In addition, land consolidation facilitates the interaction of the agricultural industry with other industries (e.g., ecological tourism and service industry) and the formation of featured rural brands. As shown in Fig. 1, earlier studies have shown that land consolidation contributes to shaping rural industrial development through industrial productivity (Huang et al., 2011), industrial extension (Long and Qu, 2018), industrial intersection (Xiong, 2021), and industrial features (Kong et al., 2019).

First, land productivity refers to the improvement of agricultural production efficiency. Land consolidation improves the productivity through the scale development of land use, enhancing the quality of cultivated land and ameliorating agricultural production conditions (Long and Qu, 2018). Land fragmentation is alleviated through the concentration of plots (Li et al., 2018), which has significant impacts on the scale of agricultural production (Long, 2014). This increases the quantity of cultivated land. Second, industrial extension refers to the extended growth of the agricultural industry. As a form of agricultural reform, land consolidation contributes to the modernization of farms and their structures (Colombo and Perujo-Villanueva, 2019). Consolidated land can be used to facilitate the development of modern agriculture as well as the organic forest and fruit industries (Zhou et al., 2020), which extends the value added to the agricultural industry and increases the proportion of agriculture to GDP growth (Huang et al., 2011). Third, industrial intersection refers to the integrated

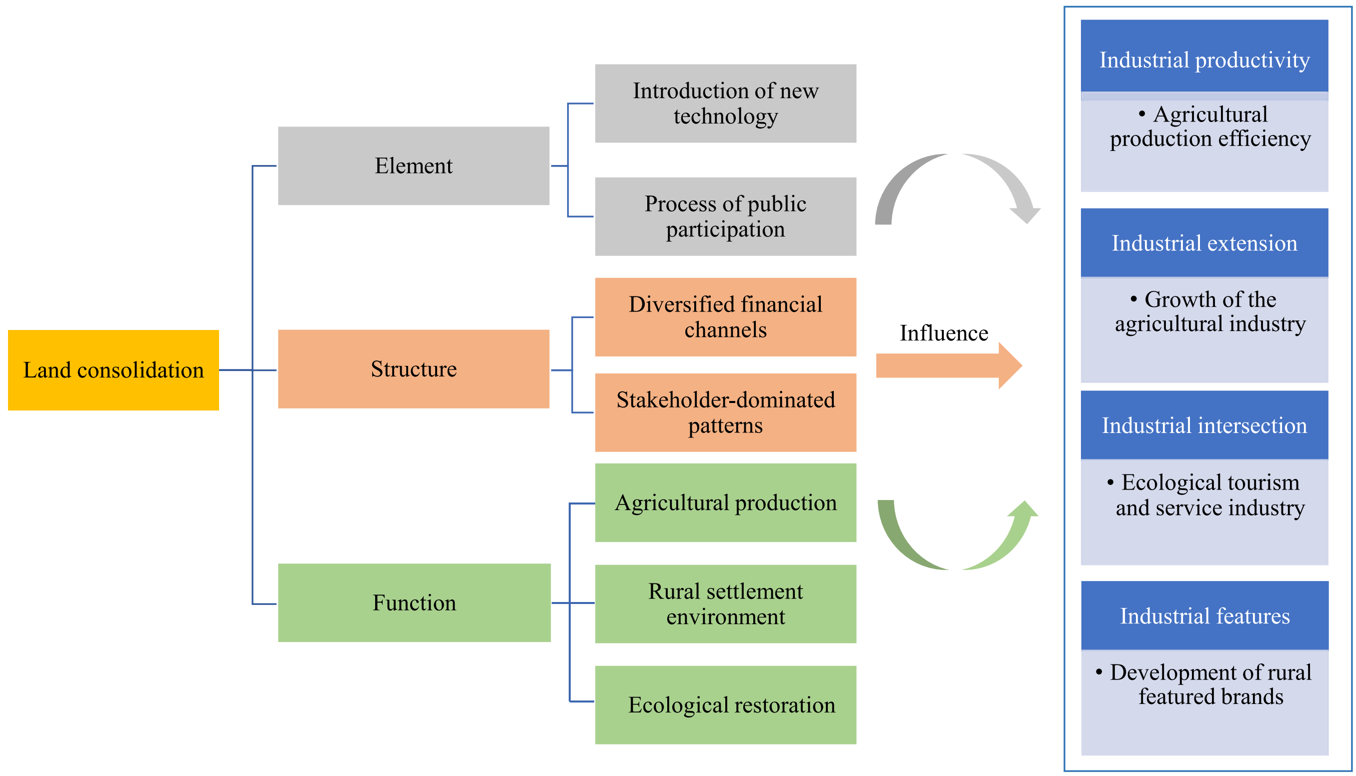


Fig. 1. Influencing mechanism of land consolidation on industrial development.

development of agriculture with other industries, such as the ecological tourism and service industry. Land consolidation promotes the development of rural ecological tourism and service industry in the spatial restructuring of rural areas by enhancing regional accessibility and building ecological landscape corridors (Xiong, 2021). Fourth, the aspect of industrial features refers to the development of featured rural brands. Rural revitalization requires a distinctive development path. Building featured brands increases the competitiveness of products and services in rural areas (Zhou et al., 2020). Villages with unique landscape features, advantageous locations, and distinctive historical cultures can be tapped as part of featured brands and developed through land consolidation, such as improving rural infrastructures and realizing aesthetic values of rural cultivated land and housing (Long et al., 2019).

The implementation of land consolidation is a complex system involving a variety of factors, such as technology introduction, financing structure, public engagement, and functional layout (Jiang et al., 2022; Jürgenson, 2016). These factors produce complex and non-linear effects on rural industrial development. However, there is still a dearth of research examining the relationship between land consolidation and rural development, and in particular no comprehensive exploration has been undertaken incorporating both qualitative and quantitative analyses of this relationship. To achieve the desired outcome (e.g., the expected level of industrial development), factors related to land consolidation need be reasonably managed. On the one hand, these factors are not independent, and there are interactions between them. This study, therefore, explored how different configurations of factors related to land consolidation shape industrial development. On the other hand, it is necessary to determine to what extent these factors constrain (or contribute to) the different levels of industrial development, shedding new light on the nuanced effects of each factor. By this means, this study aims to form an in-depth and systematic understanding of the relationship between land consolidation and rural industrial development.

3. Research design

3.1. Methods

In this study, both QCA and NCA are employed to create a systematic understanding of how and to what extent land consolidation contributes to industrial development. First, QCA is a comparative multi-case-driven analysis method that combines both qualitative and quantitative approaches (Invernizzi et al., 2020). This method integrates theoretical perspectives and case knowledge to investigate the characteristics of multiple cases (Gavetti and Levinthal, 2000). QCA assumes that the impact of case outcomes may not be the consequence of a single factor but rather a complex interaction among a range of factors (Qin et al., 2021; G. Wang et al., 2021; W. Wang et al., 2021). A key aspect of QCA is its use of Boolean algebra to analyze truth tables (Greckhamer et al., 2018). A truth table contains all logically possible configurations (i.e., different combinations of related factors that lead to a similar outcome) (Rihoux and Ragin, 2009). Second, necessary conditions are also an important aspect of forming a systematic understanding of a phenomenon (e.g., the occurrence of a specific outcome). A necessary condition is what enables an outcome to exist (Dul et al., 2020). That is, without the necessary condition, the outcome cannot exist. QCA can examine the necessity, but this is conducted separately from the sufficiency analysis (Schneider and Wagemann, 2012). As suggested by Fainshmidt et al. (2020), NCA is an effective method to identify the necessity. Traditional regression-based methods identify conditions that (on average) contribute to an outcome, but NCA can identify necessary (critical) conditions that would prevent a certain outcome from occurring (i.e., constraints or bottlenecks) (Dul, 2016). More specifically, NCA shows how different conditional factors constrain the achievement of a certain outcome (Dul, 2016). NCA sets a ceiling for the data, representing the level of a conditional factor that is necessary for a given outcome (Dul et al., 2020). The results of QCA show the different combinations of conditional factors that are sufficient to explain a certain outcome. NCA complements QCA and further reveals the nuanced effects of each conditional factor on different levels of the outcome. As such, through the lens of element-structure-function, this study combines QCA and NCA to explore the complex and non-linear relationship between land consolidation and rural industrial development. Research processes and outcomes are shown in Fig. 2.

3.2. Case and data

In this study, multiple cases were collected that met the following four criteria. First, the research cases derive from formal and authoritative channels, including a collection of typical cases of land consolidation for promoting poverty alleviation (China Land Press) and the China Land Consolidation website (http://www.lcrc.org.cn/). The selected cases are representative and reflect the typical land consolidation projects being performed in different regions of China. The scale of the case analysis is the county, as land consolidation usually involves a range of relevant projects (e.g., connectivity infrastructures) that conducted within the county, rather than being limited to one specific village or another. Second, case selection follows the most similar conditions, different outcome (MSDO) or most different conditions, similar outcome (MDSO) criterion (Rihoux and Ragin, 2009). As such, no additional control variables were added because MSDO and MDSO isolate the variations caused by other factors through cross-case comparisons (Held and Gerrits, 2019). Third, Ragin and Strand (2008) indicate that it is crucial to understand the actual situation of each case. In this study, the potential completeness of data sources was taken into consideration in the case selection. Fourth, the cases were all complete by 2018 because the Ministry of Natural Resources of China issued an important pilot announcement for comprehensive land consolidation in 2019. Comprehensive land consolidation has achieved significant benefits in coastal areas (especially in Zhejiang Province). However, in central China and other relatively underdeveloped areas, the implementation of comprehensive land consolidation has not produced the expected results and faces a series of problems resulting from technical, public, financing, organizational, production, living, and ecological factors (Pan et al., 2023). This study not only sheds light on the ongoing pilot of comprehensive land consolidation but also provides guidance for rural industrial development. Using these criteria, 32 cases were selected for analysis. Table 1 and Fig. 3 show the distribution of these cases and their land consolidation projects. Finally, drawing on the number of cases selected, QCA-based case studies can be divided into three categories, including small sample, medium sample, and large sample. In this study, the number of selected cases fits the category of medium sample analysis (10–40 cases) and typically involves four to eight conditional factors (Invernizzi et al., 2020). Consequently, we select seven conditional factors after a systematic analysis of the selected 32 cases.

3.3. Variable measurement

3.3.1. Outcome variables measurement

Industrial development seeks to achieve broader economic benefits through the integration of land, products, people, and markets (Cao et al., 2020). As indicated in Table 2, this study selected four indicators to assess outcome variables, i.e., industrial productivity, industrial extension, industrial intersection, and industrial features. A comparison of similar cases shows that differences in the outcome can be explained by analyzing differences among the conditional factors (Held and Gerrits, 2019). QCA analysis provides an effective way to isolate other factors that affect rural industrial development and focuses on the configurational effect of factors related to land consolidation. The selected four outcome variables are closely related to land consolidation projects. First, the industrial extension helps diversify rural industries. Due to the decrease in the added value of the primary industry in terms of GDP, the industrial chain expands into secondary and tertiary industries. Second, industrial integration in rural development is usually reflected in the integration of agricultural and service industries. Tourism is one of the largest parts of the service industry in rural areas. Land consolidation improves the landscape and creates favorable conditions for the development of rural tourism (Bonadonna et al., 2020; Wang and Li, 2019). Third, land productivity reflects industrial productivity in agriculture, which is closely related to the implementation of land consolidation. Fourth, a number of branded industries take advantage of local resources. Through land consolidation, the local agricultural industry can achieve scale, standardized, and market-oriented development, eventually forming leading products and branded industries. For example, Tongxin County conducted a series of projects during land consolidation, such as high-standard cultivated land construction projects, agricultural construction projects, forestry construction projects, environmental remediation projects, and transportation construction projects. After the intensive land consolidation, the agricultural productivity of Tongxin County was significantly improved. Another typical example can be seen in Xinchang County. Xinchang County launched a series of land consolidation projects to promote rural revitalization. In particular, it introduced social capital (e.g., in the form of a cultural tourism development firm) to develop the tea tourism industry through land consolidation projects and form a series of tourist areas (e.g., Xiayanbei is rated as a 3 A tourism village in Zhejiang Province).

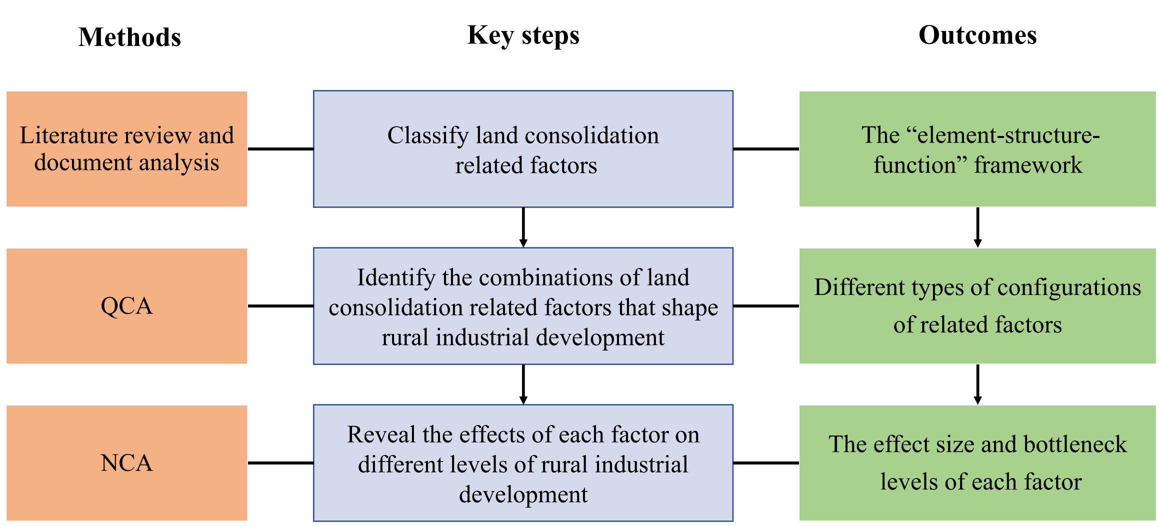


Fig. 2. Research process and outcomes.

Table 1
The selected land consolidation cases.

Case number	Province	County	Typical project	Case number	Province	County	Typical project
C1	Anhui	Huoqiu	Panji Township land consolidation project	C17	Inner Mongolia	Jalaid Banner	Dry-land conversion to irrigated land consolidation project
C2	Anhui	Mengcheng	Major land consolidation project in cooperation between Xiaojian Town and Tancheng Town	C18	Jiangsu	Lianshui	Hongkao Town land consolidation project
C3	Fujian	Minqing	Shanglian Town land consolidation project	C19	Jiangsu	Sihong	Taiping Town land consolidation project
C4	Fujian	Qingliu	Laifang Town land consolidation project	C20	Jiangxi	Nanchang	Jiangxiang Town land consolidation project
C5	Guangxi	Lingshan	Lingcheng Tow land consolidation project	C21	Ningxia	Haiyuan	Sanhe Town land consolidation project
C6	Guizhou	Hezhang	“Boost economy and benefit people” major land consolidation project	C22	Ningxia	Pingluo	Xijing Town land consolidation project
C7	Guizhou	Tongzi	“Boost economy and benefit people” major land consolidation project	C23	Ningxia	Tongxin	Xinglong Town land consolidation project
C8	Hainan	Lingao	Jialai Town land consolidation project	C24	Shandong	Dongping	Xinhu Town land consolidation project
C9	Hebei	Xingtang	Oral Town land consolidation project	C25	Shandong	Pingyin	Meigui Town, Kongcun Town, and Ancheng Town land consolidation projects
C10	Heilongjiang	Longjiang	Shanquan Town land consolidation project	C26	Shanxi	Wuqi	Baibao Town land consolidation project
C11	Henan	Jia	Comprehensive agricultural development land consolidation project	C29	Sichuan	Xuyong	Zhengdong Town land consolidation project
C12	Henan	Xichuan	Major land consolidation projects for soil transfer and fertilization	C28	Sichuan	Yanting	Jurong Town land consolidation project
C13	Henan	Taikang	Banqiao Town and Sunkou Town land consolidation projects	C27	Yunnan	Luoping	Laochang Town land consolidation project
C14	Hubei	Jiayu	Panjiawan Town land consolidation project	C30	Yunnan	Songming	Xiaojie Town land consolidation project
C15	Hunan	Fenghuang	Sangongqiao Town land consolidation project	C31	Zhejiang	Kaihua	Linshan Town land consolidation project
C16	Hunan	Sangzhi	Ruitapu Town and Liujiaping Town land consolidation projects	C32	Zhejiang	Xinchang	Yulin Street land consolidation project

Note: The basic information and data of selected cases come from the case collection published by China Land Press and the China Land Consolidation website (http://www.lcrc.org.cn/); C=case.

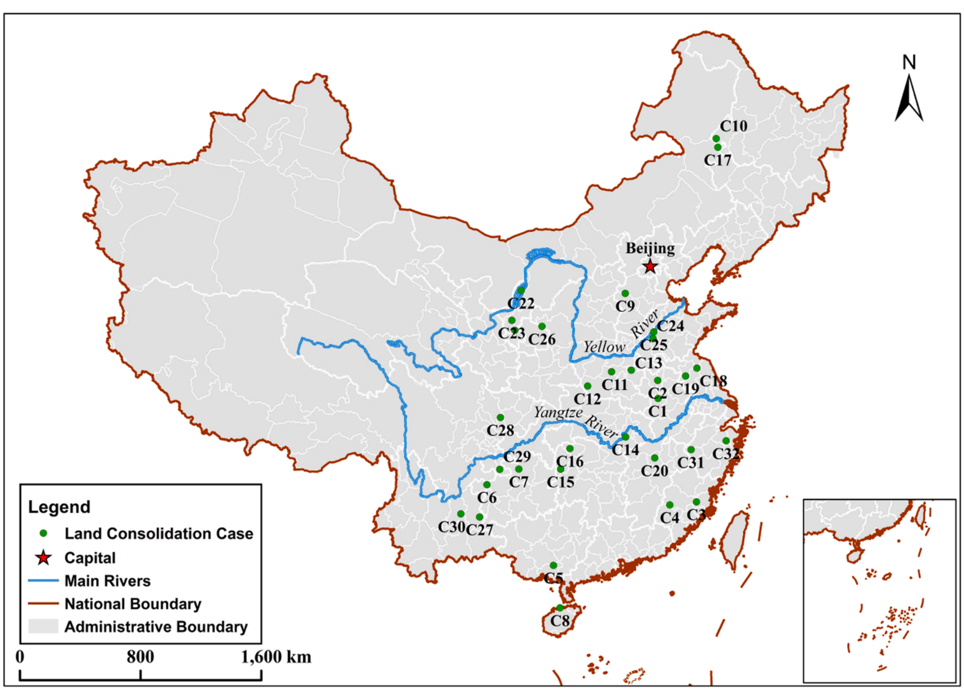


Fig. 3. The distribution of selected land consolidation cases.

3.3.2. Conditions variables measurement

Elements, structures, and functions are three indispensable parts of a system. Different configurations of elements shape the structures and functions of the system. First, as a complex and systematic tool, land consolidation involves public and technical elements, such as public participation and technological introduction. Second, land consolidation usually requires an intensive investment. Some projects are entirely dependent on government investments. An increasing number of

Table 2
The measurement indicators of outcome variables.

Outcome variables	Symbol	Measurement indicators	Sources
Industrial extension	ie	Primary industry value added (% of GDP)	Long and Qu (2018)
Industrial integration	ii	Number of tourist areas	Xiong (2021)
Industrial productivity	ip	Land productivity	Huang et al. (2011)
Industrial features	if	Number of branded industries	Kong et al. (2019)

Note: The Drcnet statistical database system (http://www.drcnet.com.cn/www/int/) and Cabinet statistical database (https://db.cei.cn/) supplement the data for outcome variables.

projects are turning to rely on the market investment, especially due to the rise of PPP financing. According to the available financial structure, organizational structures reveal the power of coordination between the government and the market. Specifically, land consolidation can be divided into two organizational models, i.e., government-led model and market-led model (Xie et al., 2021). The government-dominated model of land consolidation forms a top-down approach that relies on the leading power of the government. In this model, the government's natural resources sector is the main implementer and investor in land consolidation projects, and also encourages the involvement of market resources, such as providing rewards after project completion (instead of subsidies before project implementation). The market-dominated model is a bottom-up approach that leveraging market resources, such as professional cooperatives, agricultural firms, and family farms. Third, land consolidation manifests a range of functions in restructuring production, living, and ecological spaces. Table 3 shows the measurement indicators of conditional variables in this study. It should be noted that counties are characterized by similarities and aggregations at a given time (e.g., in their institutional environment, development goals, and land consolidation strategies). This study assigns conditional variables based on the typical land consolidation projects collected from public sources within a specific county at a given time (i.e., as of 2018) and uses these projects as a proxy to reflect the status of holistic land consolidation.

4. Results

4.1. QCA results

4.1.1. Necessity analysis

QCA is an emerging method that combines both qualitative and quantitative analyses and allows the identification of a series of related factors jointly explaining an outcome (Popp et al., 2021) and hence can be used to reveal the complex effects of land consolidation on rural industrial development. A calibration procedure was performed to standardize the data. This process converts non-standard data into values between 0 and 1 (G. Wang et al., 2021; W. Wang et al., 2021). Following Fiss (2011) and Greckhamer (2016), this study uses direct calibration method and sets three fixed-value anchor points: full membership (fuzzy score = 0.95), crossover point (fuzzy score = 0.5), and full non-membership (fuzzy score = 0.05). For example, as the outcome variable, 1 refers to the highest level of rural industrial development (full membership), 0.5 refers to a moderate level (neither high nor low), and 0 refers to the lowest level (Pot et al., 2019). A descriptive statistical analysis of the sample data was conducted to identify extreme values (i.e., maximum and minimum) and the mean values of all variables (Table 4). The mean value was then converted into a fuzzy score of 0.5, and the maximum and minimum values were converted to 0.95 and 0.05, respectively (G. Wang et al., 2021).

In QCA analysis, consistency describes the effect of a conditional factor on a certain outcome. A conditional factor is considered as a necessary condition when its consistency is greater than 0.9 (Ragin,

Table 4
The results of descriptive statistics of calibrated variables.

Variables	Mean	Standard deviation	Minimum	Maximum
te	0.64	0.43	0.05	0.95
fs	0.44	0.45	0.05	0.95
pe	0.44	0.32	0.05	0.95
os	0.56	0.45	0.05	0.95
pf	0.64	0.27	0.05	0.95
lf	0.50	0.32	0.05	0.95
ef	0.36	0.27	0.05	0.95
Industrial development	0.50	0.30	0.05	0.95

Note: te= technical element; pe= public element; fs= financing structure; os=organizational structure; pf= production function; lf= living function; ef= ecological function.

Table 3
The measurement indicators of conditional variables.

Conditional variables		Symbol	Measurement indicators		Source
Element	Technical element	te	Already introduced	1	Liu (2018)
			Not introduced	0	
	Public element	pe	Planning	No less than three stages assigned to 1	Wang et al. (2019)
			Implementation	Otherwise, 0	
			Completion and handover		
			Operation and maintenance		
Structure	Financing structure	fs	Government investment	0	W. Wang et al. (2021)
			Multi-party investment (including private investment)	1	
	Organizational structure	os	Government-led model (top-down approach)	0	Xie et al. (2021)
			Non-government-led model (bottom-up approach)	1	
Function	Production function	pf	High-standard cultivated land consolidation	Cumulative assignment	Yang et al. (2020)
			Abandoned industrial and mining land reclamation		
			Agricultural infrastructure development		
	Living function	lf	Residential improvement		Long (2014)
			Rural landscape improvement		
	Ecological function	ef	Shelterbelt development		Li et al. (2019)
			Soil reconstruction		

Note: the county statistical yearbook of the National Bureau of Statistics of China supplements the data for conditional factors (i.e., land consolidation); [b] If one aspect of the indicator system is met, a value of 1 is assigned; if two aspects of the indicator system are met, the value is assigned as 2, and so on (An et al., 2017); te= technical element; pe= public element; fs=financing structure; os=organizational structure; pf=production function; lf=living function; ef=ecological function.

2006). As shown in Table 5, no conditional factor reaches the consistency of 0.9, which indicates that the necessity condition does not exist. NCA further reveals the extent to which a conditional factor constrains the occurrence of an outcome.

4.1.2. Configuration analysis

QCA results indicate that three types of configurations are sufficient for explaining a high level of rural industrial development, i.e., function-driven configuration (Type 1), function-element-driven configuration (including Type 2a and Type 2b), and function-structure-driven configuration (Type 3); the consistencies of these configurations are 0.847, 1.000, 0.860, and 0.993, respectively. In Table 6, a filled circle indicates the existence of a condition, a crossed circle indicates the absence of a condition, and a blank indicates a fuzzy status (i.e., the condition may be present or absent) (Seny Kan et al., 2016). Furthermore, the size of the circle is used to distinguish between core and peripheral conditions. Larger circles represent core conditions that may have a significant impact on the outcome, and small circles represent peripheral conditions, which produce an auxiliary impact on the outcome (Fiss, 2011).

As shown in Table 6, the production function (pf) is a core condition for each type of configuration. The production function underpins the basic features of cultivated land. The enhancement of the production function through land consolidation can effectively increase land capacity, reduce costs in agricultural production, and ultimately contribute to poverty alleviation and industrial development. Therefore, the production function of cultivated land is the underlying aspect of land consolidation that shapes rural industrial development. The first configuration is the function-driven type. The raw coverage of this configuration is 0.250, indicating that it features the greatest explanatory power for predicting the outcome. As shown in Table 6, the function-driven type involves the core presence of the production function and the peripheral presence of the technical element and organizational structure. Specifically, this configuration indicates that market-led land consolidation leverages appropriate technological tools to improve land production function and ultimately contributes to improving rural industrial development. For example, in Nanchang County, a series of land consolidation projects has encouraged the participation of professional cooperatives, agricultural firms, and family farms by providing incentives upon the project completion (instead of subsidies prior to project implementation). Nanchang County has established a comprehensive monitoring platform that utilizes the Internet of Things to achieve the dynamic supervision of land consolidation. Overall, the land consolidation in Nanchang County has proven to be effective, especially in agricultural development. The function-driven type improves land production with the use of market power and technological tools. Market-led land consolidation contributes to the efficient use of resources and reduces the underlying problems associated with inadequate public participation in land consolidation (Zhang et al., 2019).

The second configuration is the function-element-driven type, which includes Type 2a and Type 2b. The raw coverages of Type 2a and Type 2b are 0.239 and 0.194, respectively. This type of configuration involves playing the essential role of elements in improving functions. As shown in Table 6, Type 2a involves the core presence of the production function and the peripheral presence of the technical element, public element, financing structure, and living function. Specifically, this configuration shows that the adoption of appropriate technological tools, a high level of public participation, and multi-party investment are beneficial for the improvement of land production and living function, ultimately contributing to rural industrial development. For example, in Sihong County, a series of land consolidation projects have promoted the application of information technology and driven public participation throughout the land consolidation process. In addition to government support, the financial source for land consolidation also involves the private sector. Leveraging information technology tools, effective public participation, and diversified financial sources, Sihong County enables an effective improvement of land production and living function. Type 2b involves the core presence of the production function and the peripheral presence of the technical element, public element, and living function. Unlike Type 2a, Type 2b emphasizes the adoption of a top-down approach, which suggests a government-led land consolidation model. For example, Mengcheng County has set up a government leadership group to tackle tough challenges during the implementation process of land consolidation projects.

The third configuration is the function-structure-driven type. The raw coverage of this configuration was 0.182. As shown in Table 6, this type involves the core presence of the production function and the peripheral presence of the public element, organizational structure, financing structure, and ecological function. In this configuration, both two structure factors play an important role in integrating the element and function factors. This configuration indicates that a market-led model with a high level of public participation and multi-party investment contributes to improving the production and ecological functions

Table 6
The comprehensive analysis of configurations.

Conditions		Configurations			
		Function-driven Type 1	Function-element-driven		Function-structure-driven Type 3
			Type 2a	Type 2b	
Element	te	•	•	•	⊗
	pe	⊗	•	•	•
Structure	os	•		⊗	•
	fs	⊗	•		•
Function	pf	⬤	⬤	⬤	⬤
	lf		•	•	⊗
	ef	⊗	⊗	⊗	•
Raw coverage		0.250	0.239	0.194	0.182
Consistency		0.847	1.000	0.860	0.993

Note: ⬤ indicates the presence of a core condition; • indicates the presence of a peripheral condition; ⊗ indicates the absence of a peripheral condition; ⨂ indicates the absence of a core condition; blank indicates that the condition can be either present or absent; te= technical element; pe= public element; fs= financing structure; os=organizational structure; pf= production function; lf= living function; ef= ecological function.

Table 5
The analysis of necessary conditions of QCA.

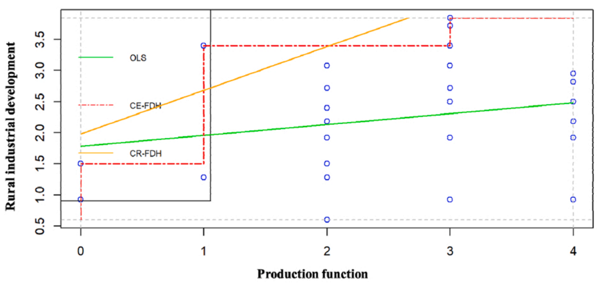
Conditional variables	Consistency	Coverage	Conditional variables	Consistency	Coverage
te	0.817	0.642	~ te	0.282	0.396
pe	0.581	0.669	~ pe	0.581	0.669
fs	0.697	0.791	~ fs	0.403	0.365
os	0.823	0.746	~ os	0.276	0.313
pf	0.850	0.669	~ pf	0.473	0.662
lf	0.669	0.674	~ lf	0.634	0.640
ef	0.561	0.787	~ ef	0.757	0.595

Note: ~ represents the negation which refers to the low level (absence) of outcomes. te= technical element; pe= public element; fs= financing structure; os=organizational structure; pf= production function; lf= living function; ef= ecological function.

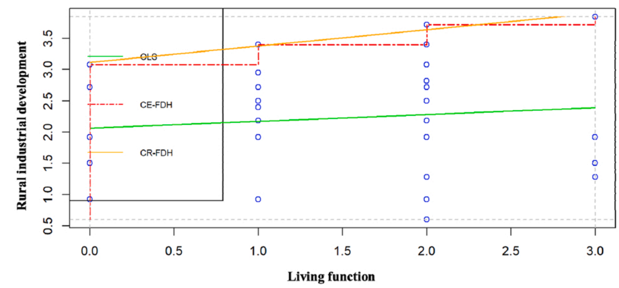
and ultimately facilitates rural industrial development. For example, in Hezhang County, land consolidation projects are administered by the government's natural resources department and agricultural firms, which absorb the local workforce to carry out projects and improve the ecosystem of cultivated land.

4.2. NCA result

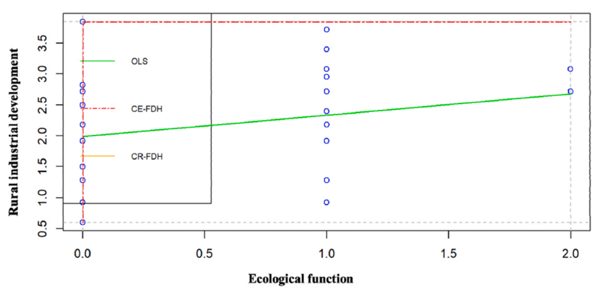
NCA is another emerging method that can examine the necessary but not sufficient effects of different factors and show how these factors constrain an outcome to occur (i.e., bottlenecks) (Jabeur, 2020). Three aspects are involved in NCA analysis, including scatter plots, effect sizes, and bottlenecks (Dul, 2016). A scatter plot represents of the overall



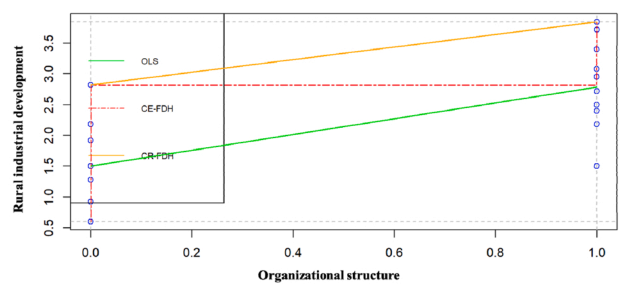
(a) Production function and rural industrial development.



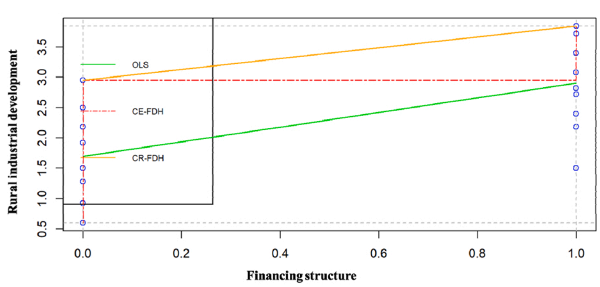
(b) Living function and rural industrial development.



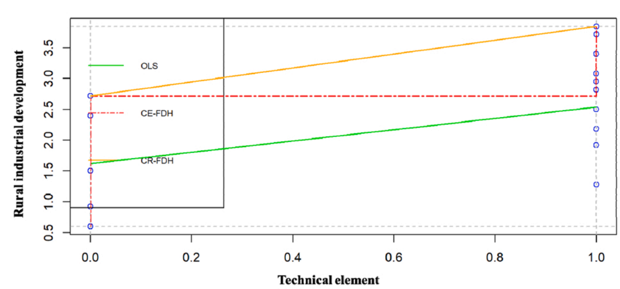
(c) Ecological function and rural industrial development.



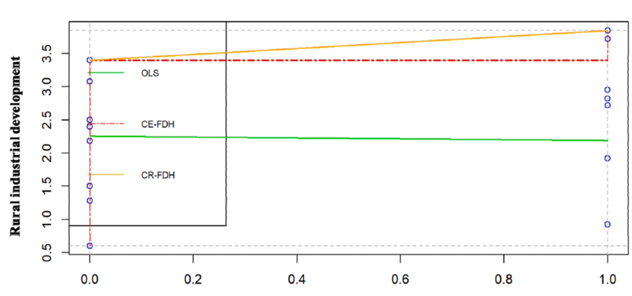
(d) Organizational structure and rural industrial development.



(e) Financial structure and rural industrial development.



(f) Technical element and rural industrial development.



(g) Public element and rural industrial development.

Fig. 4. Scatter plots of conditional variables and rural industrial development.

relationship between conditions (X) and the outcome (Y) (Dul et al., 2020). The effect size refers to the degree to wich conditions (X) constrain the outcome (Y) (Dul, 2016). A bottleneck shows the marginal level (critical value) for a given effect (Dul et al., 2020).

First, an empty zone appears in the upper left corner of the scatter plot, which is termed the ceiling zone (Goertz et al., 2013); this zone demonstrates the necessary conditions (X) for the outcome (Y) (Dul et al., 2020). More specifically, this empty zone shows that a high value for Y is not possible with low values of X, suggesting that X constrains Y. As shown in Fig. 4, traditional data analysis of the relationship between factors related to land consolidation and rural industrial development involves an ordinary least squares (OLS) regression line passing through the middle zone (green line). The ceiling line indicates that there is a ceiling on rural industrial development depending on the factors related to land consolidation. The two ceiling lines refer to ceiling envelopment with free disposal hull (CE-FDH) and ceiling regression with free disposal hull (CR-FDH) (Dul et al., 2020). CE-FDH is the default for a nonparametric approach (red line), and CR-FDH is the default for a parametric linear approach (yellow line). The ceiling line separates the zone where no observations are available and defines the size of the ceiling zone (Dul, 2016). CE-FDH is preferred when a straight ceiling line does not properly represent the data along the zone border. The relationships between different factors related to land consolidation and rural industrial development are shown in Fig. 4.

Second, the effect size of the necessary condition depends on the size of the ceiling zone. The effect is stronger if the ceiling zone is larger (Dul, 2016) and the effect size is expressed as d. Specifically, $0 < d < 0.1$ is considered a small effect, $0.1 \leq d < 0.3$ is a medium effect, $0.3 \leq d < 0.5$ is a large effect, and $d \geq 0.5$ is a very large effect (Dul, 2016; Dul et al., 2020). In NCA, two analysis methods, i.e., ceiling envelopment (CE) and ceiling regression (CR) are usually applied (Dul, 2016). In this study, both methods were used to test the robustness of the results. As indicated in Table 7, the effect size of the ecological function determined by two methods small ($d<0.1$). Therefore, the ecological function demonstrates a relatively weak effect on rural industrial development. Apart from the ecological function, other conditional factors demonstrate medium or large effects.

Third, the bottleneck table is another way of exhibiting ceiling lines. As shown in Table 8, the observed values (ranging between 0 and 1) represent the threshold levels of conditional factors (X) for achieving the desired level of the outcome (Y) (Jabeur, 2020). In QCA analysis, all conditional and outcome variables are calibrated. Table 8 shows the diversified roles of conditional factors (i.e., factors related to land consolidation) to different outcome levels (i.e., rural industrial development). The ecological function is not a necessary condition for different levels of rural industrial development. As the level of rural industrial development approaches 0.5, the production function becomes a necessary condition. When the level of rural industrial development reaches 0.7, the technical element and organizational structure become necessary conditions. When the level of rural industrial development reaches 0.8, the public element, living function, and financing structure become necessary conditions. To reach a high level of rural industrial development (90% quantile), the production function needs to reach 0.549, the living function 0.516, the technical element 0.712, the financing structure 0.639, the public element 0.516, and the organizational structure 0.684. For achieving the highest level of rural industrial development, the requirements for different conditional factors become significantly elevated, such as public element (from 0.516 to 0.928), financial structure (from 0.639 to 1.000), organizational structure (from 0.684 to 1.000), living function (from 0.516 to 0.928), and technical element (from 0.712 to 1.000).

Table 7
Analysis outcomes of necessary conditions of NCA.

Conditions	Method	Accuracy	Ceiling zone	Effect size (d)
te	CE	100.0%	1.130	0.348
	CR	100.0%	0.565	0.174
pe	CE	100.0%	0.388	0.138
	CR	90.6%	0.255	0.104
fs	CE	100.0%	0.900	0.277
	CR	100.0%	0.450	0.138
os	CE	100.0%	1.030	0.317
	CR	100.0%	0.515	0.158
pf	CE	100.0%	0.813	0.250
	CR	96.9%	0.622	0.191
lf	CE	100.0%	0.450	0.138
	CR	93.8%	0.340	0.104
ef	CE	100.0%	0.000	0.000
	CR	100.0%	0.000	0.000

Note: CE = ceiling envelopment. CR = ceiling regression. The accuracy is defined as the number of observations at or below the ceiling line divided by the total number of observations, multiplied by 100%; te= technical element; pe= public element; fs= financing structure; os=organizational structure; pf= production function; lf= living function; ef= ecological function.

Table 8
The bottleneck table of the NCA analysis.

Y Industrial development level	X1 te	X2 pe	X3 fs	X4 os	X5 pf	X6 lf	X7 ef
0	NN	NN	NN	NN	NN	NN	NN
0.1	NN	NN	NN	NN	NN	NN	NN
0.2	NN	NN	NN	NN	NN	NN	NN
0.3	NN	NN	NN	NN	NN	NN	NN
0.4	NN	NN	NN	NN	NN	NN	NN
0.5	NN	NN	NN	NN	0.088	NN	NN
0.6	NN	NN	NN	NN	0.203	NN	NN
0.7	0.137	NN	NN	0.053	0.319	NN	NN
0.8	0.425	0.104	0.278	0.369	0.434	0.103	NN
0.9	0.712	0.516	0.639	0.684	0.549	0.516	NN
1	1.000	0.928	1.000	1.000	0.665	0.928	NN

Note: NN = not necessary; te= technical element; pe= public element; fs= financing structure; os=organizational structure; pf= production function; lf= living function; ef= ecological function.

5. Discussion

5.1. Theoretical implications

Land consolidation measures are being developed and extensively implemented worldwide (Janus and Ertunç, 2020) to cope with a series of problems, such as inefficiencies in production, obstacles to agricultural modernization, reductions in biodiversity, and soil erosion (Colombo and Perujo-Villanueva, 2019; Janus and Ertunç, 2021). However, with the development of society and economy, the connotations of land consolidation (e.g., objectives, tasks, and requirements) are evolving, which alters the focus from parcel rearrangement to poverty alleviation and rural revitalization, especially in developing countries that are challenged by rural poverty (Zhou et al., 2020). Earlier qualitative studies summarized the roles of land consolidation in facilitating rural development by integrating three types of factors (i.e., element, structure, and function) (Li et al., 2018; Long et al., 2019; Zhou et al., 2020), such as promoting technology adoption (Wang and Li, 2019), extending the public involvement (Li et al., 2019), and establishing a diversified financing mechanism (Li et al., 2018). In addition, other quantitative studies have mainly focused on the effect of a specific factor related to land consolidation, such as organizational structure (Xie et al., 2021), production function (Wojewodzic et al., 2021), and public element (Liu et al., 2016). Rural industrial development provides endogenous power for rural revitalization (Lu et al., 2021). However, as land consolidation is an important platform for integrating resources to promote rural revitalization, there is a paucity of quantitative research that assesses and characterizes the role of land consolidation in promoting rural industrial development in a systematic and holistic approach (Rao, 2022). Taking into account the underlying causal complexity in management research (Misangyi et al., 2016), this study applies QCA to analyze the effects of different configurations of factors related to land consolidation on rural industrial development. Furthermore, NCA is used to assess to what extent various factors constrain different levels of rural industrial development. In this way, this study provides a series of theoretical contributions to the land consolidation literature. In addition, the configurational approach of this study may profoundly influence the methodological paradigm in the field of land consolidation, facilitating a shift from symmetric to asymmetric perspectives in data analysis.

First, this study reveals three types of configurations of factors related to land consolidation in promoting rural industrial development, including function-driven configuration, function-element-driven configuration, and function-structure-driven configuration. Sustainable rural industrial development necessitates the implementation and optimization of land consolidation measures using these three types of configurations. To improve the effectiveness of rural industrial development, it is necessary to perform land consolidations from a systematic perspective and promote an appropriate combination of element-structure-function factors. The three configurations represent a nonlinear system of seven factors of land consolidation, including technical element, public element, financing structure, organizational structure, production function, living function, and ecological function. This configuration echoes previous calls, such as Long et al. (2019), Zhou et al. (2019), and Li et al. (2018), to reveal the governance ecosystem of land consolidation in promoting rural revitalization. The production function is present in all three types of configurations, which plays a pivotal role in promoting rural industrial development. In land consolidation, the production function appears to work together with living and ecological functions, but a full understanding of how they can combine when a high level of industrial development is reached also requires the element and structure factors to be taken into account. Specifically, land consolidation facilitates the restructuring of rural production spaces, such as improving agricultural productive conditions and achieving a transformation of large-scale agricultural production. Therefore, land consolidation provides a carrier and motivation for rural industrial development. Element, structure, and function, the three types of conditional factors are interdependent and jointly affect rural industrial development (Long et al., 2019). This study extends the theoretical lens for assessing the effectiveness of land consolidation (Wojewodzic et al., 2021) and provides a systematic perspective reflecting complex mechanisms for leveraging land consolidation to promote rural industrial development (Rao, 2022).

Second, rural industrial development is constrained to varying degrees by a series of conditional factors. NCA analysis provides a nuanced and in-depth understanding of the diversified roles of factors related to land consolidation, which reveals how necessary they are for achieving different levels of rural industrial development. This insight advances the body of knowledge on these three types of conditional factors that generally, although not always, shape rural industrial development (Long et al., 2019; Zhang et al., 2020; Zhou et al., 2020). The production function is the basic necessary condition that underpins rural industrial development, which constrains the medium level of development (50% quantile). The technical element, financing structure, and organizational structure become necessary conditions at the upper-medium levels of rural industrial development (between 50% and 80% quantile). To achieve a high level of industrial development (80% quantile), the public element and living function become necessary conditions. Public involvement and living conditions have been shown to be crucial for the process of maximizing the value of land consolidation (Jiang et al., 2022; Krupowicz et al., 2020). The results of NCA shed additional light on the threshold of achieving a high level of rual industrial development through effective public involvement.

This study also makes a relevant methodological contribution to lines of research in land consolidation and rural revitalization, as it adopts a mixed qualitative and quantitative method embracing causal complexity and capturing nuanced results. First, this study echoes the call to move beyond the multiple regression method (Misangyi et al., 2016; Woodside, 2013), as the influence of the conditional factor on the outcome can differ, depending on the other factors that are included in the analysis. Most causal relationships are asymmetrical. A given conditional factor does not always lead to the same outcome. Thus, this study offers new directions and insights for future research by applying emerging methodological approaches and producing intriguing empirical results regarding the impact of land consolidation on rural industrial development. Second, QCA is a promising research method in the field of land consolidation because of its applicability to situations where the number of available cases is in the range of 10–50 (Invernizzi et al., 2020). These case sizes are common in the analysis of land consolidation projects. More particularly, QCA is expected to generate significant insights by promoting cross-case comparisons (Wang et al., 2023). Case data could be qualitative and/or quantitative (Invernizzi et al., 2020). Third, in response to the suggestion of Fainshmidt et al. (2020), this study applies NCA to validate and extend the results of QCA. The examination of necessity promotes a substantially different understanding of the relationship between conditional factors and the outcome. NCA demonstrates sufficient factors to yield a given outcome. A necessary factor allows an outcome to exist. Conversely, if the factor does not reach an appropriate level, the outcome is not likely to occur. Thus, this study provides a novel methodological perspective with its introduction of NCA as an additional logic and data analysis toolkit for an in-depth understanding of factors related to land consolidation and rural industrial development.

5.2. Policy implications

Land consolidation has been an important part of land use policy (Janus and Ertunç, 2021). In China, land consolidation forms a strategic tool to coordinate population, land, and industry and plays a crucial role in integrating element-structure-function factors (Long et al., 2019), thereby promoting industrial development and rural revitalization (Xie et al., 2020; Zhou et al., 2020; Zhang et al., 2020). Taking the differences in the element-structure-function factors into account, the driving mechanisms of rural industrial development through land consolidation demonstrate distinctive characteristics. The results of QCA and NCA indicate three policy suggestions to align strategies for promoting rural industrial development.

First, function-driven configuration is suitable for rural areas that lack autonomy and rely mainly on government investments. These areas should consider developing diversified rural economic organizations. Governments can provide rewards after the project completion (instead of providing subsidies before the project implementation) to encourage rural self-governance and boost endogenous development. For example, the government of Longzhou County specified the relevant incentives for land consolidation projects led mainly by professional cooperatives, agricultural firms, and family farms. These incentives have greatly mobilized public participation and crowdfunding in land consolidation. Land consolidation relying on government leadership often suffers from failures to meet the actual needs of farmers (Zhang et al., 2019). Public involvement in economic and social activities is instrumental for improving the rural atmosphere. Family farms, agricultural firms, and cooperatives develop from the increased autonomy of farmers. In 2019, the Ministry of Natural Resources issued its notice on the pilot project of comprehensive land consolidation, which encourages the involvement of collective organizations and farmers, as well as enabling the public to participate in the planning and design, construction, and operation and maintenance stages of land consolidation. The results of this study show that the presence of market-led models (i.e., family farms, agricultural firms, and cooperatives) are important conditions when rural industries reach a certain level of development. Specifically, when rural industry reaches an upper medium level of development (70% quantile), land consolidation should not only guarantee the basic production function but also, more importantly, explore bottom-up organizational approaches and introduce new technologies. In this type of configuration, the organizational structure acts as the determinant for shaping rural industrial development. The extension of farmers' autonomy entails making land consolidation more transparent and better able to meet the needs of rural development and livelihoods.

Second, function-element-driven configuration (including Type 2a and Type 2b) is appropriate for rural areas that place reduced emphasis on the ecological function of the land. These areas are anticipated to exploit the potential of public participation, enhance the development of technology, and achieve a restructuring of living spaces. In Ganzi County, the government launched a series of lectures to advocate for land consolidation. Land consolidation promotes the excavation of local characteristics, which conduces to attracting social capital and stimulating the endogenous power of industrial development. The restructuring of rural living spaces optimizes the relationship between people and land, taking advantage of the ecological endowment (Tu et al., 2018). When rural industry reaches a high level of development (80% quantile), land consolidation is increasingly dependent on effective public participation. In this case, a crowdsourcing approach (Krupowicz et al., 2020), gathering a combination of public knowledge and resources across the life-cycle of land consolidation projects, is especially useful for achieving a high-quality transformation of rural industries.

Third, function-structure-driven configuration corresponds to rural areas with technical restrictions and medium quality living conditions. These areas are expected to enhance their grassroots leadership and develop forward-looking planning to further protect the ecological function at the front end of land consolidation projects. With environmental problems becoming serious, the ecological function is prominent in the process of economic and social development. For example, Jinxiang County developed a land consolidation plan (2016–2020) that emphasizes the improvement of the ecological environment, including geological disaster management, pollution prevention, and the promotion of ecological tourism. Currently, the ecological protection and restoration of rural areas have been identified as one of the focused tasks of land consolidation.

5.3. Limitations and future research directions

This study combines QCA and NCA to explore the nuanced influencing mechanisms of land consolidation on rural industrial development. Despite its contributions, this study has limitations that call for future research. First, QCA is a multiple-case comparative analysis method that combines qualitative and quantitative approaches. Future research could focus on a specific region, for example, the central region of China, to conduct a longitudinal and qualitative single case study. Second, this study selected 32 typical land-consolidation cases. The sample size of QCA analysis is usually between 15 and 80 cases (Hanckel et al., 2021). Therefore, 32 cases may only be a moderate sample size. Future research could further expand the sample. Third, the cases reviewed in this study were all from China. So far as the relationship between land consolidation and rural development is concerned, more and more studies are being conducted in a single country, such as the Netherlands (Stańczuk-Gałwiaczek et al., 2018), Slovakia (Muchová and Raškovič, 2020), Albania (Zhllima et al., 2021), Poland (Wojewodzic et al., 2021), India (Munnangi et al., 2020), Iran (Asimeh et al., 2020), and China (Zhou et al., 2020, 2019). The reason for focusing on a specific country is that mixed cases in different institutional environments may weaken the explanatory power of conditional factors (Wang et al., 2018). Therefore, a future expansion could develop series of parallel analyses in different countries and then compare them separately.

6. Conclusion

Land is the crucial carrier of rural vitalization (He et al., 2020). Land consolidation has become an important and effective tool to shape the relationships among population, land, and industry (Cao et al., 2020). As a complex system, land consolidation manifests as the configurations of element, structure, and function factors. Different configurations yield various effects for rural industrial development. Through the lens of the element-structure-function framework, this study reveals how the configurations of factors related to land consolidation shape rural industrial development based on QCA analysis. This study also investigates to what extent these factors constrain rural industrial development based on NCA analysis. The concluding remarks are presented as follows:

Different land consolidation-related factors (i.e., technical element, public element, financing structure, organizational structure, production function, living function, and ecological function) are interdependent and jointly shape rural industrial development. Land consolidation is a comprehensive project that calls for systematic thinking to examine the influencing mechanisms of conditional factors, and provides insights for rural revitalization through a series of project-based measures in agricultural land preservation, built-up land regeneration, and ecological conservation. In different configurations, these factors play various roles. The production function occurs in all three types of configurations. Over the course of facilitating rural industrial development, it is necessary to take the production function of land as the core driving linkage for activating other conditions. Based on the role of core and peripheral conditions, three typical configurations can be categorized: type 1 is the function-driven configuration, type 2 is the function-element-driven configuration, and type 3 is the function-structure-driven configuration. In the process of promoting land consolidation, the approach of focusing on deficient conditional factors faces limitations. Thus, it is necessary to formulate targeted policies based on three types of configurations as a whole, especially taking into account other conditional factors related to deficient factors. First, type 1 configuration relates to rural areas that are less autonomous and depend mainly on government investments. For type 1, it is necessary to focus on activating the technical element and developing rural economic organizations (especially new business entities such as family farms) in the process of land consolidation. Through the introduction of technology, type 1 encourages those with a willingness to farm on a long-term basis to expand the scale of their operations. On this basis, several family farms can further join together to form farmers' cooperatives to provide integrated operation services. In addition, the government's investment is powerful to stimulate the endogenous development of rural areas through the way of rewards after the completion of land consolidation projects (instead of subsidies before project implementation). Second, type 2 configuration relate to rural areas with less emphasis on the land ecological function. For type 2, there is a need to focus on exploring the potential of public participation, enhancing the introduction of technologies, and improving living conditions. In particular, the restructuring of living spaces contributes to reconciling the relationship between people and land to drive the human-nature-coupled system in sustainable development. Third, type 3 configuration relates to rural areas that face technological constraints and disadvantaged living conditions. For type 3, it is suggested to reinforce the participation of the public at different project stages to better activate their potential as non-government parties, especially enhancing the protection of the land ecological function at the front end of land consolidation projects.

Furthermore, the progressive rural development with differentiated policy priority can be induced based on the nuanced role of the production function, living function, technical element, public element, organizational structure, and financing structure, as revealed by NCA analysis. On this basis, this study classifies the level of rural industrial development into three stages. The first stage corresponds to the lower-middle development level (below 50% quantile). No necessary conditions occur. The second stage corresponds to the upper-middle development level (between 50% and 80% quantile). The production function, technical element, financing structure, and organizational structure gradually become necessary conditions for rural industrial development. The third stage corresponds to the high development level (above 80% quantile). The public element and living function become necessary conditions. To achieve a high-quality rural industrial development, effective public involvement across different land consolidation stages (i.e., planning, implementation, completion and handover, and operation and maintenance) becomes the crucial condition. Meanwhile, the improvement of living conditions is critical to the value maximization of land consolidation. NCA analysis discovers and demonstrates the necessity for attaining a certain level of the outcome, which provides the basis for formulating targeted land consolidation policies based on different levels of rural industrial development. To better drive rural industrial development, it is effective to combine element-structure-function factors based on three types of configurations (i.e., function-driven, function-element-driven, and function-structure-driven configurations). Furthermore, at different stages of rural industrial development, the potential of land consolidation can be unlocked based on the bottleneck levels of each factor.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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